INTERNAL MANAGEMENT REVIEW

FMR Operations v3

Comprehensive Project Documentation

Management overview, user procedures, technical architecture, deployment, operations, governance, and readiness assessment

Management position The repository demonstrates a capable, controlled FMR operating platform and is suitable for a limited production pilot after the Priority 0 controls in this document are completed. It should not yet be represented as an unrestricted enterprise production release while it remains alpha software.

Prepared for	Project and Operations Management
Prepared by	Jonathan Muratalla
Document date	August 3, 2026
Repository	jonaDaniM/FMRv3
Reviewed release	3.0.0-alpha.19
Source snapshot	main branch, commit 52d5f902e1e70dd1477dbe93e348a06eec2a3d04

Classification: Internal working document. Validate production configuration, security scope, ownership, and operating procedures before broad distribution.

Document Control

Item	Value
Document version	1.0
Status	Management Review Draft
Review scope	Current main-branch source, release history, manifests, data contracts, portal views, core services, and operational controls
Review exclusions	Live Google Workspace permissions, deployed production data, actual user activity, cloud audit logs, quota consumption, and end-user acceptance evidence were not independently tested
Update trigger	New core library release, database schema change, security model change, production deployment, or material workflow change

Contents

How to use this document Executives can read Sections 1-3 and 16-17. Field, material-control, and system users should use Sections 4-7 and 13. Technical owners should use Sections 8-15 and the appendices.

1. Executive Summary

FMR Operations v3 is a Google Workspace-based material-request control platform designed to move Field Material Requests from intake through field fulfillment while preserving traceability, role separation, and operational recoverability.

The system combines a browser-based portal with a versioned Google Apps Script library and a structured Google Sheets data store. The current repository snapshot is release 3.0.0-alpha.19. It contains separate Field, Admin, and Owner workspaces; searchable published FMRs; material locating, reserving, issuing, and backorder workflows; controlled staging and publication; bulk workbook ingestion; user administration; test/production environment binding; audit records; health monitoring; backups; and guided recovery.

1.1 Management conclusion

Recommended decision Authorize a controlled operational pilot after completing the Priority 0 production controls: deployment-scope review, production environment provisioning, formal access approval, backup/recovery validation, test evidence, and an approved operating owner/support model.

Assessment area	Current assessment	Management implication
Business fit	Strong	The workflow directly addresses material-request visibility, physical availability, reservation, issuance, and exceptions.
Functional coverage	Broad for an alpha release	Field, Admin, Owner, import, operational-readiness, and lifecycle capabilities are represented in the codebase.
Control design	Above average for a rapid Apps Script solution	Roles, environment identity, read-only mode, locks, indexes, audit, backups, and recovery reduce operational risk.
Technology maturity	Controlled pilot / alpha	The release label, large source modules, limited repository governance, and platform constraints require disciplined rollout.
Production readiness	Conditional	Code-based safeguards exist; live security, Workspace sharing, quotas, and recovery evidence still require formal validation.
Strategic value	High if adoption is managed	The platform can replace fragmented spreadsheets and provide a unified operational record without major infrastructure cost.

1.2 What management is receiving

- **A single operating model:** One lifecycle for intake, publication, field activity, exceptions, and closure.
- **Management visibility:** Published FMR register, fulfillment quantities, active reservations, pending backorders, health status, and backup status.
- **Separation of duties:** Read Only, Field, Material Admin, and System Owner profiles.
- **Controlled data entry:** Manual staging plus validated bulk import from Excel or Google Sheets.
- **Recoverability:** Health checks, backup history, scheduled operations, recovery previews, and auditable corrective actions.
- **A defined path to production:** Priority-based remediation and a practical go-live checklist are included in this document.

2. Management Overview and Decision Brief

2.1 Problem addressed

Field material operations often break down when requested quantities are treated as physically available, when the same material is informally reserved for multiple users, when partial issuance is not reflected against the original requirement, or when backorders are recorded without an approval trail. FMR Operations v3 establishes explicit states and controlled transactions so the project can distinguish what was requested, what was located, what is reserved, what remains available, what was issued, and what requires exception handling.

2.2 Business outcomes enabled

Outcome	How the system supports it	Suggested KPI
Faster field response	Indexed search by official FMR or ISO and sheet; material-line actions are generated from current state.	Median time from search to transaction
Reduced double allocation	Bag & Tag records reserve material and remove it from general availability.	Conflicting allocation incidents
Better exception control	Field submits backorder requests; Admin confirms, rejects, or returns them.	Pending backorder age; return rate
Improved accountability	Authenticated email, performer, recipient, timestamp, correlation ID, and audit information are recorded.	Transactions with complete metadata
More reliable reporting	Header totals aggregate line-level states and the Admin register exposes operational filters.	Fulfillment percent; remaining requirement
Improved continuity	Health history, backups, scheduled checks, and controlled recovery actions are included.	Backup age; health-pass rate; recovery test success

2.3 Management decision gates

Gate	Evidence required	Owner	Decision
Security scope	Confirmed deployment audience, Google Workspace sharing, least-privilege OAuth and embedding decision.	Project IT / Security	Approve or remediate
Production data boundary	Dedicated production spreadsheet and backup folder, environment fingerprints, restricted sharing.	System Owner / IT	Approve
Operational acceptance	Signed Field, Admin, and Owner scenarios using representative FMRs.	Material Control / Field	Approve
Recovery readiness	Successful backup, restore/reconcile rehearsal, recorded RTO/RPO.	System Owner / IT	Approve
Support model	Named owner, deputy, incident contact, release process, and change calendar.	Project Management	Approve
Pilot metrics	Baseline and target KPIs, review date, and stop/go criteria.	Project Controls	Approve

3. Project Purpose, Scope, and Business Value

3.1 Purpose statement

The purpose of FMR Operations v3 is to provide a controlled system of record for Field Material Requests associated with installation work packages, ISO drawings, material line items, physical availability, bagging/tagging, field issuance, and backorder review. The system is intended to preserve a one-to-many relationship between a work package, its ISO/sheet combinations, FMRs, and material lines while allowing multiple FMRs to exist for the same ISO/sheet without incorrectly combining them.

3.2 In scope

- Authenticated portal access through the user's Google account.
- Field search by official FMR number or combined ISO and sheet.
- Material actions: confirm available, Bag & Tag, locate and issue, issue available, issue from a bag, and submit backorder.
- Admin dashboard, published-FMR register, operational filters, backorder decisions, and active Bag & Tag visibility.
- Owner-controlled manual staging, validation, publication, and FMR renumbering.
- Bulk intake from approved Excel workbooks or Google Sheets, with worksheet-level FMR parsing and issue review.
- Role management, test/production environment controls, maintenance/read-only mode, and configuration defaults.
- Operational health checks, backup retention, scheduled operations, controlled recovery, audit history, diagnostics, acceptance fixtures, and migrations.

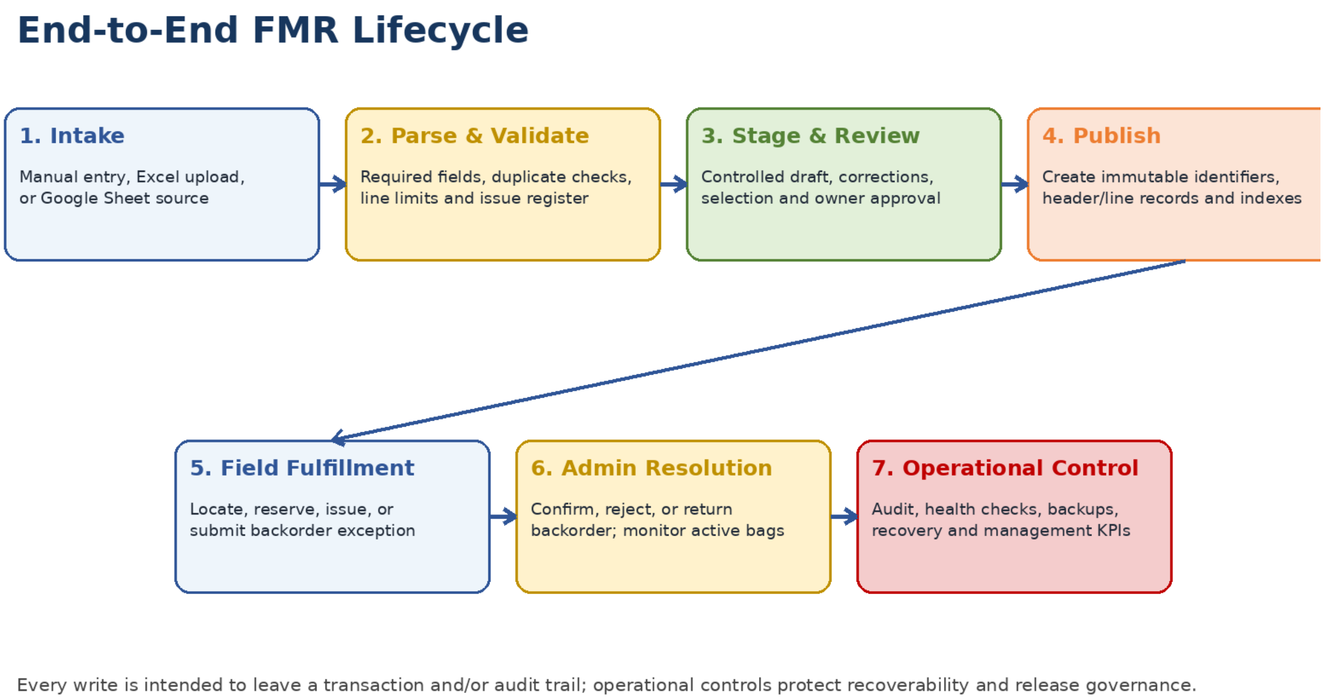
3.3 Not demonstrated or not currently represented as a complete product feature

- Native mobile applications or offline operation.
- Enterprise identity federation beyond authenticated Google accounts and the internal Users register.
- Automated integration to ERP, procurement, warehouse-management, schedule, or document-control systems.
- Electronic signatures, badge scanning, barcode/QR scanning, or hardware label printing as a confirmed end-to-end feature.
- A conventional relational database, external API gateway, centralized secrets vault, or enterprise observability platform.
- Formal CI/CD workflows, automated unit-test framework, semantic release notes, or repository contribution governance in the reviewed tree.
- Independent penetration testing, live load testing, or legal/compliance certification.

3.4 Primary stakeholders

Stakeholder	Interest	Primary system interaction
Field personnel	Find the correct FMR and record actual material disposition quickly.	Field view
Material Control / Admin	Resolve shortages, review pending exceptions, and monitor reserved material.	Admin view
System Owner	Control master data, users, environments, intake, release, backups, and recovery.	Owner view
Project Controls / Planning	Use quantities, status, age, and throughput for coordination and reporting.	Admin reports / exports
Project Management	Approve governance, monitor KPIs, and decide rollout scope.	Management review
IT / Security	Validate access, deployment, supportability, data protection, and continuity.	Apps Script / Workspace administration

4. End-to-End Business Process



4.1 Lifecycle narrative

Intake. The System Owner creates a draft manually or begins a bulk import from an Excel workbook or Google Sheet. In the bulk model, each worksheet represents one proposed official FMR.

Validation. The system checks required header values, material lines, quantities, ISO/sheet information, duplicate conditions, line limits, and source-format issues. Problems are recorded as errors, warnings, or informational issues.

Staging. Valid or correctable records are stored in staging. The Owner can inspect, edit, and select items before publication. Staging separates source-data preparation from the live operational register.

Publication. Publication creates stable FMR and line identifiers, initializes quantity states, writes search-index entries, records the source staging relationship, and changes the draft to published.

Field fulfillment. Field users search the official record and perform only actions currently allowed by line state. Transactions update line quantities, header totals, Bag & Tag records, notices, and audit history.

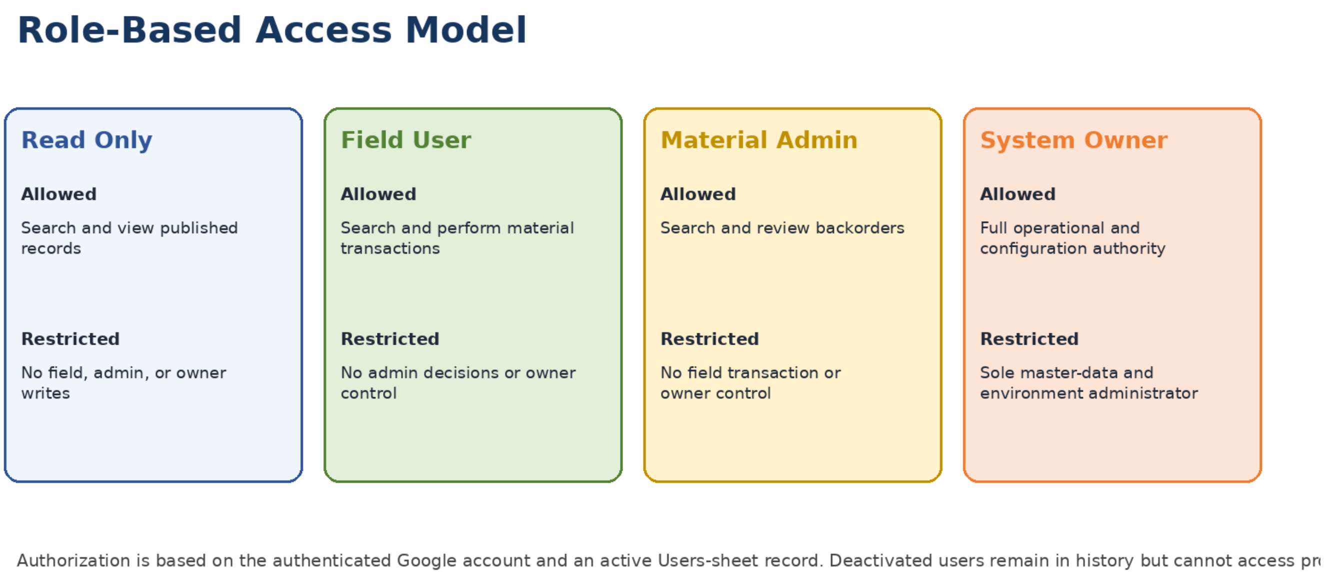
Exception handling. Field users request a backorder rather than confirming one. Material Admin users review the request and may confirm all or part, reject it, or return it for field review.

Operational control. The Owner monitors health, backup age, stale exceptions, stale bags, recent activity, and environment identity. Controlled recovery can refresh totals, rebuild indexes, synchronize notices, or perform safe reconciliation.

4.2 State and accountability principles

- **Requested is not received.** Requested quantity represents demand. It does not become located, available, reserved, or issued without a controlled field transaction.
- **Reservation is exclusive.** Bagged material is tracked in Bag & Tag records and is not treated as generally available until issued or otherwise administratively resolved.
- **Issuance requires a recipient.** Issue actions require the name of the recipient and preserve the authenticated account and performer metadata.
- **Backorder is a two-step control.** Field reports the exception; Admin determines whether it is confirmed, rejected, or returned.
- **Aggregates derive from lines.** FMR header totals reflect the material-line state and are refreshed after transactions.
- **Every correction must be explainable.** Correlation IDs, transactions, audit records, and recovery records support traceability.

5. Users, Roles, and Responsibilities



5.1 Permission matrix

Capability	Read Only	Field	Admin	Owner
Search published FMRs	Yes	Yes	Yes	Yes
Perform field transactions	No	Yes	No	Yes
Review backorders	No	No	Yes	Yes
Create/edit staging	No	No	No	Yes
Publish or renumber FMRs	No	No	No	Yes
Manage users/configuration	No	No	No	Yes
Change environment or transaction mode	No	No	No	Yes
Run backup/recovery operations	No	No	No	Yes

5.2 Responsibility definitions

Role	Responsibilities	Control boundary
Read Only	Look up records and support coordination without changing operational state.	No write permission.
Field User	Record where material was found, reserve it, issue it to a recipient, and submit unresolved quantities for review.	Cannot make admin decisions or change master data.
Material Admin	Monitor operations, review backorders, and determine the disposition of shortages.	Cannot perform field transactions or change owner configuration.

Role	Responsibilities	Control boundary
System Owner	Administer users, defaults, environments, intake, publication, renumbering, backups, health, and recovery.	Owner authority should be limited, named, and periodically reviewed.

5.3 User lifecycle

1. **Request access.** Manager identifies the required profile and business reason.
2. **Owner creates or updates the user.** Enter the Google account email, display name, profile, and notes.
3. **User signs in.** The active/effective Google email must match one active Users record.
4. **System records access.** Last login and interface information are updated where supported.
5. **Periodic review.** Owner and management review active users and role appropriateness.
6. **Deactivate, do not erase.** Inactive users remain in history but are denied access; record the reason for deactivation.

> **Separation-of-duties recommendation** For production, avoid assigning Owner to routine field users. Use at least one primary Owner and one approved recovery deputy, with management approval for temporary elevation.

6. Nontechnical User Guide

6.1 Access and navigation

1. Open the approved FMR Operations v3 web-app URL using the authorized Google account.
2. Choose Field, Admin, or Owner from the top navigation. Views are visible according to the page route, but protected server actions still enforce the user's profile.
3. Confirm that the page subtitle shows the expected user, role, core version, and environment.
4. When the system is in READ_ONLY mode, read the maintenance message and do not attempt transactional work until the Owner restores ENABLED mode.

> **Access error** An "Unauthorized user" or missing-account message normally means the signed-in Google email does not match an active Users entry. Contact the System Owner; do not share credentials.

6.2 Field workflow - search

1. **Choose a search method.** Use Auto Detect for most searches, FMR Number for an exact official number, or ISO + Sheet for an ISO-specific lookup.
2. **Enter the value.** Examples: an official FMR number, or a combined ISO and sheet such as FG-70912_001|3. The search also recognizes a slash or "SHT/SHEET" style in supported cases.
3. **Review the result card.** Confirm FMR, IWP, requester, status, and the material line before acting.
4. **Check the quantity strip.** Requested, located, bagged, available, issued, pending backorder, confirmed backorder, and remaining quantities must be interpreted separately.
5. **Use only the action presented.** The system generates allowed actions from current state. A missing action usually means the quantity is unavailable, reserved, locked by a pending review, or complete.

6.3 Field action procedures

Action	When to use and required discipline
Confirm Available	Use when material has been physically located but is not being reserved or issued immediately. Enter quantity, storage location, and performer. Do not use this action based only on the requested quantity.
Bag & Tag	Use to reserve located material for the FMR. Enter quantity, storage location, and performer. Record notes that help identify the physical bag/tag when needed.
Locate & Issue	Use when material is found and immediately delivered. Enter quantity, recipient, performer, and optional location/notes. The recipient is required.
Issue Available	Use for material already recorded as available. Enter quantity, recipient, performer, and notes as needed.
Issue From Bag	Select the correct active Bag & Tag source, then enter quantity, recipient, and performer. Never issue from a tag associated with a different physical package.
Submit Backorder	Use when the unresolved requirement cannot be located. Enter quantity, reason, performer, and useful field notes. This is a request for Admin review, not a confirmed backorder.

> **Field transaction checklist** Before submitting, verify the FMR/ISO/sheet and physical material; enter only the actual quantity handled, location, and recipient when issuing; add useful exception notes; then confirm the updated quantities on screen.

6.5 Admin workflow

1. **Refresh the dashboard.** Review published FMR count, active tags, backorder queue, and operational rail.
2. **Use the FMR register.** Filter by search type, query, status, priority, or operational exception. Sort by activity, date required, FMR number, remaining quantity, fulfillment, or requested quantity.
3. **Open one FMR.** The register intentionally loads header summaries first; opening a record retrieves its indexed line details.
4. **Review pending backorders.** Validate the FMR line, requested exception quantity, field reason, notes, and current material situation.
5. **Choose a decision.** Confirm all or part of the pending quantity, reject it, or return it for field review. Record decision notes.
6. **Monitor Active Bag & Tags.** Review reserved and partially issued tags, age, location, and remaining quantities. Escalate stale or questionable reservations.
7. **Recheck results.** Confirm that the queue, line state, and header totals reflect the decision.

6.6 Backorder decision guide

Decision	Use when	Result
Confirm	The shortage is accepted as a true backorder. Partial confirmation is allowed within the unresolved quantity.	Moves quantity from pending to confirmed backorder; the request may remain partially pending.
Reject	The backorder request is not valid or material should be handled through another field action.	Clears the rejected pending quantity and creates field-facing resolution information where applicable.
Return	Field must recheck, clarify, or act before Admin can decide.	Creates or preserves a returned-for-review condition; partially confirmed requests may split the returned remainder.

Admin control Do not confirm a backorder merely to clear the queue. The decision changes the operational record and should be supported by material-control knowledge and useful notes.

6.7 Owner workflow - system controls

1. Refresh Production Controls and verify project name, environment, database fingerprint, transaction mode, and user summary.
2. Create or update authorized users using the minimum profile required.
3. Use READ_ONLY mode for maintenance, investigation, or deployment holds. Enter a clear maintenance message.
4. Keep default craft, destination, delivery contact, priority, and timezone current.
5. Before changing environments, confirm the target database is explicitly configured and its logical environment matches the binding.
6. After any access or configuration change, validate with a separate user account representing the affected role.

6.8 Owner workflow - manual intake and publication

1. Create a new staging record or open an existing draft.
2. Enter official FMR number when known, IWP number, requester, date required, priority, notes, and at least one material line.
3. For every line, provide ISO number, ISO sheet, quantity greater than zero, UOM, and the identifying material information available.
4. Save the draft and correct all validation errors.
5. Before publication, confirm the official FMR number is correct and not already published.
6. Publish only after review. Publication creates live operational records and indexed search entries.
7. Use renumbering only for a legitimate correction, with a reason, and verify downstream references afterward.

6.9 Owner workflow - Bulk Import Wizard

1. **Choose source.** Upload an approved Excel workbook within the configured limits or provide an accessible Google Sheet reference.
2. **Start parsing.** The system creates a batch and evaluates each worksheet as one proposed official FMR.
3. **Review batch summary.** Check proposed FMR count, total line count, valid items, warnings, errors, and parser status.
4. **Inspect issues.** Correct blocking errors such as missing FMR, IWP, ISO/sheet, material headers, invalid quantities, duplicates, or line-limit problems.
5. **Resolve controlled overrides.** Apply an ISO-sheet override only when source evidence supports it and the confirmation requirement is satisfied.
6. **Select items.** Stage only the intended valid/warning items. Do not stage blocked items.
7. **Review staging.** Bulk import does not bypass owner review or publication. Inspect the resulting staged FMRs before making them live.
8. **Retain source traceability.** Preserve batch ID, source filename, source fingerprint, and issue history for later investigation.

> **Fractional sizes** Release alpha.19 explicitly protects Size columns as plain text to avoid spreadsheet conversion of values such as 1/2 or 3/4 into dates or decimals. Source workbooks should still be reviewed for prior coercion.

6.10 Owner workflow - Operational Readiness

1. Refresh the Operations Center and review overall health, integrity, schema, user control, backup age, stale backorders, stale bags, and recent transactions.
2. Run a manual health check after deployments, schema changes, imports, or suspected data issues.
3. Create a manual backup before high-risk work and record useful notes.
4. Install one approved daily operations trigger at the project’s selected hour and verify timezone.
5. Use recovery preview before applying a recovery action.
6. Apply only the minimum recovery needed: refresh header totals, rebuild the search index, synchronize field notices, or safe reconcile.
7. Review the Recovery Actions and Audit Log afterward and communicate material changes to operations.

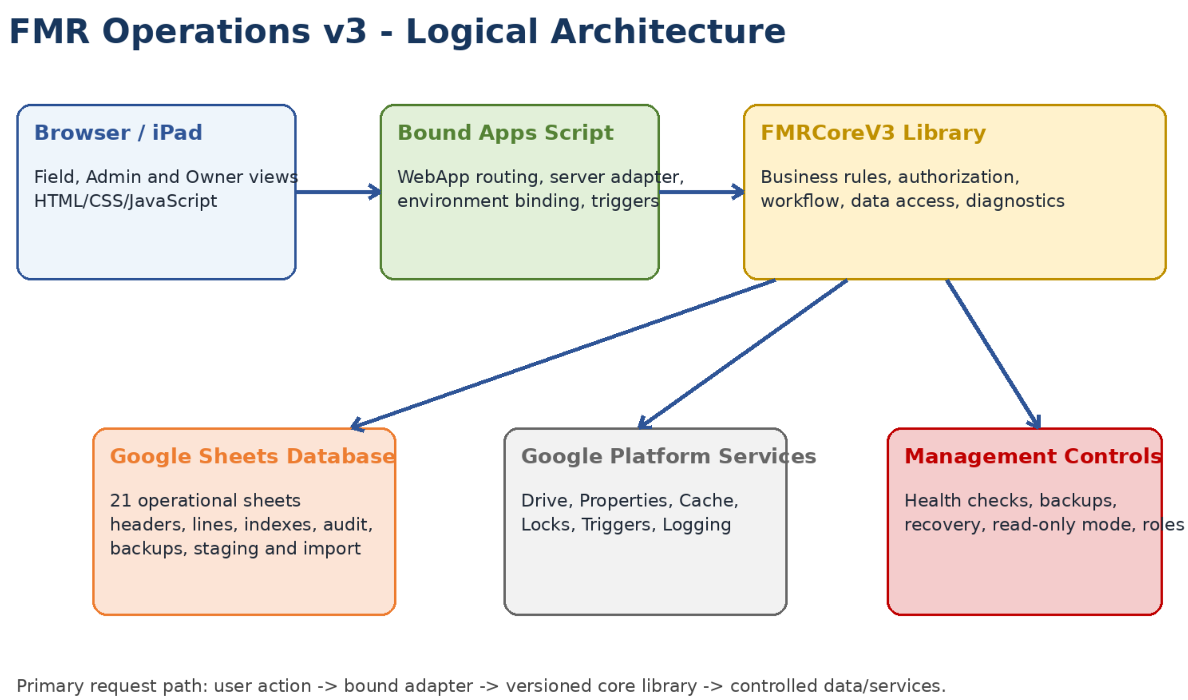
7. Functional Capability Reference

Capability	Behavior represented in repository
Portal routing	Three routes: Field, Admin, Owner; shared protected services and common version/environment bootstrap.
Search	Exact indexed FMR, ISO/sheet, and line lookups; auto-detection and admin contains-search fallback.
Field state engine	Calculates permitted actions and workflow phase from requested, located, bagged, available, issued, backorder, and remaining values.
Material transactions	Records transaction type, quantity, UOM, authenticated email, performer, recipient, bag references, storage, timestamp, and notes.
Bag & Tag	Creates reservation headers/items and tracks remaining quantity through partial issuance.
Backorders	Field request; Admin confirm/reject/return; supports partial confirmation and returned-remainder handling.
Admin register	Header-first pagination, exact index resolution, text fallback, status/priority/exception filters, multiple sort modes.
Staging	Create/update draft headers and lines, validation errors, superseded draft lines, source linkage.
Publication	Validates, enforces unique FMR number, creates live header/lines/indexes, and updates staging status.
Renumbering	Owner-controlled correction path with reason and downstream index handling.
User control	Four predefined profiles, activation/deactivation, access recording, owner-only administration.
Environment control	Separate TEST/PRODUCTION bindings, database fingerprint, transaction mode, maintenance message.
Bulk import	Excel/Google Sheet sources, batch/item/line/issue model, source fingerprints, limits, review and staged output.
Operational readiness	Health logs, backup history, retention, scheduled checks, recovery preview/apply, stale-item thresholds.
Diagnostics/fixtures	Schema, performance, API contracts, Field/Admin acceptance scenarios, and migration verification.

7.1 Important system invariants

- All published material lines require an ISO number, ISO sheet, positive requested quantity, and UOM.
- An official FMR number is required before publication and is checked against the search index for uniqueness.
- Field actions are quantity-limited by current line state; pending backorders may lock locatable quantity.
- Writes are wrapped in a script lock for critical field, admin, publication, and migration operations.
- Inactive users and inactive operational records are excluded from protected action paths.
- Production binding requires explicit configuration; the default database fallback applies only to TEST.
- READ_ONLY mode blocks transactional paths at the core service layer, not only in the user interface.
- Schema headers are treated as contracts and mismatches raise errors rather than silently shifting columns.

8. Technical Architecture



8.1 Architectural pattern

The repository uses a two-project Google Apps Script architecture. The Bound project is the deployable web application. It renders the portal, resolves the current Google user, binds the request to a configured TEST or PRODUCTION spreadsheet, and calls a versioned FMRCoreV3 library. The core library contains authorization, workflow, data-access, indexing, auditing, diagnostics, bulk import, and operational-readiness logic. Google Sheets functions as the operational database; Google Drive supports source files and backups; Apps Script services provide properties, caching, locks, scheduled triggers, and logging.

8.2 Component responsibilities

Component	Responsibilities	Key controls
Bound/WebApp.gs	Routes field/admin/owner requests and renders shared HTML.	Known view whitelist; server-generated URL.
Bound/Index.html, Client.html, Styles	Portal structure, client state, rendering, modal actions, user interaction.	Server authorization remains authoritative.
Bound/Adapter.gs	Identity resolution, environment binding, API wrappers, serialization, schedule installation, verification functions.	Owner validation; explicit production database; Script Properties.
FMRCoreV3/PublicApi.gs	Stable library boundary for portal operations.	Database context initialized per call; transaction gates.
Service modules	Domain logic for Field, Admin, Owner, import, notifications, operations, and controls.	Locks, validation, role checks, audit, correlation IDs.
Repository and index modules	Schema-aware row reads/writes and indexed lookup/cache.	Strict headers, exact keys, active flags, cache invalidation.
Google Sheets	Operational persistence across 21 defined sheets.	Header contracts, IDs, timestamps, active/status fields.
Google Drive / Apps Script services	File conversion/source handling, backups, configuration, caching, scheduling, logging.	OAuth scope, folder configuration, retention, audit.

8.3 Request path

1. Browser loads a view through doGet and the shared Index template.
2. Client JavaScript calls a named Bound server function through google.script.run.
3. The adapter resolves the authenticated/effective Google email and active environment.
4. The adapter supplies the database ID and user email to the versioned FMRCoreV3 library.
5. The core API initializes database context, authorizes the user, checks write mode where applicable, and invokes the domain service.
6. The repository/index layers read or write spreadsheet records and supporting Google services.
7. The server serializes the result and the client updates only the affected interface state.

8.4 Platform dependencies

Dependency	Use	Review consideration
Apps Script V8	Server-side JavaScript runtime and web-app deployment.	Execution quotas, runtime limits, release versioning.
Google Sheets API / Spreadsheet service	Primary structured data store.	Row growth, contention, formula/manual-edit governance.
Google Drive API v3 and Drive service	Source files, converted workbooks, backup copies.	Folder ownership, sharing, retention, recovery.
User Info scope	Obtain authenticated email.	Workspace account availability and privacy.
External Request scope	Permits outbound requests where used.	Confirm necessity and allowed destinations.
CacheService	User, list, configuration, and index caching.	Staleness windows and invalidation.
LockService	Serialize high-risk writes.	30-second lock wait and user retry behavior.
PropertiesService	Environment/database bindings and schedule metadata.	Owner-only access and configuration backup.
ScriptApp triggers	Daily scheduled operations.	Single trigger ownership, timezone, failure notification.
Stackdriver logging	Exception and diagnostic logging.	Retention, access, and alerting expectations.

9. Data Model and State Management

9.1 Sheet inventory

Sheet	Purpose	Domain
Dashboard	Optional spreadsheet-facing summary surface.	Derived/management
Configuration	Named settings, values, descriptions, editability.	System control
Users	Identity, role permissions, lifecycle and access fields.	Security
Lists	Controlled values such as UOM and backorder reasons.	Reference
Import_Staging_Header	Draft FMR header before publication.	Intake
Import_Staging_Lines	Draft material lines and validation results.	Intake
FMR_Header	Published FMR summary and aggregate quantities.	Operational master
FMR_Line_Items	Published material lines and state quantities.	Operational detail
Search_Index	Exact FMR, ISO/sheet, and line lookup pointers.	Performance
Operational_Index	Backorder, bag, and other queue pointers.	Performance
Material_Transactions	Append-style material activity ledger.	Audit/operations
Bag_Tag_Header	Reservation/tag identity, location, status, metadata.	Reservation
Bag_Tag_Items	Per-line reserved and issued quantities.	Reservation detail
Backorder_Requests	Field exception, admin decision, quantities, notes, status.	Exception
Audit_Log	Entity action history and before/after/payload context.	Governance
Operational_Health_Log	Health runs, statuses, counts, timings, details.	Operations
Backup_History	Backup file, environment, status, retention and errors.	Continuity
Recovery_Actions	Preview/applied recovery records and before/after JSON.	Continuity
Bulk_Import_Batches	Source-level import tracking and totals.	Bulk intake
Bulk_Import_Items	Worksheet/FMR-level parsed records and selection.	Bulk intake
Bulk_Import_Lines	Parsed material lines and source-row metadata.	Bulk intake
Bulk_Import_Issues	Error/warning/info issue register and resolution.	Data quality

9.2 Quantity model

Quantity	Meaning	Operational rule
Qty Requested	Demand stated by the FMR line.	Does not imply receipt or availability.
Qty Confirmed Located	Cumulative quantity physically located.	Increases only through locating/locate-and-issue actions.

Quantity	Meaning	Operational rule
Qty Active Bagged	Quantity reserved in active Bag & Tag records.	Excluded from general available inventory.
Qty Available	Located quantity not reserved or issued.	Can be issued or bagged within line limits.
Qty Issued	Cumulative quantity delivered to recipients.	Reduces remaining requirement.
Qty Pending Backorder	Field-reported quantity awaiting Admin disposition.	May lock new locating against the same unresolved quantity.
Qty Confirmed Backorder	Quantity accepted by Admin as backordered.	Remains tracked until fulfilled/reconciled.
Qty Not Yet Located	Unlocated portion of the original requirement.	Updated as material is located or exception state changes.
Qty Remaining Requirement	Outstanding requirement after issuance.	Primary completion measure.
Fulfillment Percent	Issued quantity relative to requested quantity.	Header summary/management KPI.

9.3 Identifiers and relationships

- FMR_ID and FMR_Line_ID are generated UUID-style internal identifiers; FMR_Number is the official business identifier.
- ISO_Key is a normalized combination of ISO Number and ISO Sheet.
- Staging IDs preserve the source-to-published lineage.
- Correlation_ID links related transaction, audit, backorder, and recovery activity.
- Bag_Tag_ID and Tag_Number identify reserved material; Bag_Tag_Items connect a reservation to a specific FMR line.
- Backorder_Request_ID identifies each field exception. Returned remainder may receive a separate request ID when required to preserve confirmed history.
- Batch_ID, Import_Item_ID, Import_Line_ID, and Import_Issue_ID preserve bulk-import traceability.

9.4 Indexing and caching

The system avoids relying exclusively on full-sheet scans for operational lookup. Search_Index stores exact FMR, ISO/sheet, and line keys with row pointers. Operational_Index stores queue-oriented keys such as backorder status and active bag status. Index results are cached with a configurable TTL, keyed by a SHA-256 digest that includes the index version. Writes invalidate the affected key. Configuration and list values use longer caches, while user authorization is cached for a shorter period.

> **Data-governance rule** Do not add, rename, reorder, or manually repurpose database columns in production. The repository validates exact header contracts and will fail on mismatches to prevent silent data corruption.

10. Module and API Reference

10.1 Bound project modules

File	Purpose
Adapter.gs	Environment/database binding, authenticated caller, core API wrappers, operations scheduling, and bound-contract verification.
WebApp.gs	doGet routing, Index template rendering, page title, and include helper.
Index.html	Field/Admin/Owner page structure and controls.
Client.html	Client state, google.script.run calls, rendering, event handlers, validation, and interaction workflows.
Styles.html	Shared visual system for portal and management/owner controls.
FieldWorkflowStyles.html	Field-specific compact workflow and action presentation.
appsscript.json	Runtime, logging, library version, deployment execution and access settings.

10.2 Core project modules

File	Purpose
Config.gs	Version, sheet names, header contracts, action constants, context, normalization, configuration cache.
Repository.gs	Schema-aware spreadsheet CRUD abstraction.
IndexServiceFmr.gs	Search/operational indexes, caching, invalidation, row pointers.
Security.gs	User lookup, authorization assertions, list cache.
PublicApi.gs	Versioned external library methods.
SearchService.gs	Search normalization, result serialization, workflow/action generation.
FieldService.gs	Material state transitions and field transactions.
FieldMetadataService.gs	Metadata validation and storage-location/notes policy.
FieldBackorderNoticeService.gs	Field-facing notifications and resolution synchronization.
BackorderService.gs	Admin queue and confirm/reject/return decisions.
AdminRegisterService.gs	Published FMR register, filters, sort, pagination.
AdminActiveBagService.gs	Active reservation queue and readiness presentation.
DashboardService.gs	Admin KPI and operational rail composition.
OwnerService.gs	Manual staging, validation, publication, renumbering.
BulkImportService.gs	Workbook/Sheet parsing, issues, batch review, staging, fractional size safeguards.
SystemControlService.gs	Profiles, user lifecycle, configuration, environment and write mode.
OperationalReadinessService.gs	Health, backup, retention, recovery, schedule support.
IntegrityService.gs	Integrity checks and reconciliation support.
AuditService.gs	Append-style audit records.

File	Purpose
Diagnostics.gs	Read-only schema/performance/API diagnostics.
Migrations.gs	Controlled data/index backfills with verification.
FieldAcceptanceFixtureService.gs	Repeatable field workflow acceptance scenarios.
AdminAcceptanceFixtureService.gs	Repeatable admin lifecycle acceptance scenarios.
appsscript.json	Advanced Drive service, OAuth scopes, V8 and logging.

10.3 Public API groups

API group	Methods	Purpose
Bootstrap and search	getFmrV3Version; getFmrV3Bootstrap; searchFmrV3	Portal initialization and indexed published-record retrieval.
Field operations	performFmrV3FieldAction	Routes validated field actions through the state engine.
Admin operations	getFmrV3AdminDashboard; getFmrV3AdminRegister; getFmrV3AdminActiveBags; reviewFmrV3Backorder	Monitoring, register, active reservations, and exception decisions.
Owner staging	saveFmrV3Staging; getFmrV3StagingList; getFmrV3StagedFmr; publishFmrV3StagedFmr	Draft lifecycle and publication.
Published corrections	renumberFmrV3; renumberFmrV3ByNumber	Owner-controlled official-number correction.
System control	getFmrV3SystemControl; saveFmrV3SystemUser; setFmrV3SystemUserActive; saveFmrV3SystemConfiguration	Access and environment configuration.
Operational readiness	getFmrV3OperationsCenter; runFmrV3OperationalHealth; createFmrV3DatabaseBackup; saveFmrV3OperationalSettings; previewFmrV3Recovery; applyFmrV3Recovery; runFmrV3ScheduledOperations	Health, continuity, and recovery.
Bulk import	startFmrV3BulkImportUpload; startFmrV3BulkImportGoogleSheet; get/update batch or item; apply ISO override; stage items; list recent batches; inspect contract	Controlled multi-FMR intake.

10.4 Service-call requirements

- Every public API call must initialize the database context before repository access.
- User email is passed from the Bound project and revalidated against the Users sheet.
- Write methods should call the core write-enabled assertion so READ_ONLY mode is enforced server-side.
- Requests should be normalized and validated in the domain service, not trusted from client JavaScript.
- Client-visible Date and Apps Script values should be serialized before returning to the browser.
- New public methods require a Bound adapter wrapper, client integration, role check, audit impact review, diagnostics, and documentation update.

11. Security, Access, and Governance

11.1 Implemented controls

Control	Implementation	Benefit
Authenticated account	Active/effective Google email is required.	Provides a named identity for authorization and audit.
Application authorization	Users sheet requires one active exact email match.	Prevents unregistered users from using protected services.
Role profiles	Read Only, Field, Admin, Owner permission flags.	Supports least privilege and separation of duties.
Owner identity	Owner action requires owner permission and configured OWNER_EMAIL match.	Restricts master-data and environment control.
Write mode	ENABLED or READ_ONLY enforced by core write paths.	Provides a controlled maintenance/incident hold.
Environment identity	TEST/PRODUCTION logical label, binding, and database fingerprint.	Reduces accidental cross-environment work.
Locks	Critical writes use ScriptLock.	Reduces concurrent quantity/state conflicts.
Validation	Required fields, state limits, uniqueness, and schema contracts.	Reduces malformed or inconsistent records.
Audit/transactions	User, action, timestamp, correlation and context records.	Improves investigation and accountability.
Backup/recovery	Backup history, retention, recovery preview/apply.	Improves continuity and controlled repair.

11.2 Security findings requiring management review

Priority	Finding	Required action
High	Web-app audience	The manifest declares web-app access as ANYONE while executing as USER_ACCESSING. Confirm whether the deployed audience is intentionally public, domain-wide, or login-gated by Google; restrict to the narrowest supported scope. Application authorization is valuable but should not substitute for appropriate deployment scope.
High	Embedding policy	The web app sets XFrameOptionsMode.ALLOWALL. Retain only if embedding is required and approved; otherwise restore restrictive framing to reduce clickjacking exposure.
High	Hardcoded identifiers	A default database ID, library ID/version, diagnostic account, fixture identifiers, and business defaults appear in source. Move environment-specific identifiers and test identities to protected configuration or separate test utilities where practical.
Medium	OAuth scope review	The core manifest includes spreadsheets, Drive, user email, and external-request scopes. Confirm every scope is necessary and document approved external destinations.

Priority	Finding	Required action
Medium	Workspace sharing	Repository code cannot prove spreadsheet, source-folder, backup-folder, or Apps Script sharing. Review owners, editors, link sharing, and transfer procedures.
Medium	Owner concentration	Owner is intentionally powerful and transfer is not exposed through the portal. Establish a controlled ownership-transfer and recovery process.
Medium	Cache revocation window	User access is cached up to 15 minutes unless invalidated by the management path. Verify deactivation invalidates cache and define emergency revocation steps.
Medium	Secrets and privacy	No secrets vault or formal data-classification controls are represented. Confirm that notes and user fields do not collect unnecessary sensitive data.
Low	Repository exposure	The repository is public. Validate that committed examples, documentation, IDs, names, and future data never disclose confidential project information.

11.3 Recommended access-governance procedure

1. Manager submits role and business need.
2. System Owner validates Google account and assigns minimum profile.
3. Owner records approver/date in notes or an external access register.
4. New user completes role-specific acceptance exercise before production use.
5. Active access is reviewed at least monthly during pilot and quarterly thereafter.
6. Terminated/transferred users are deactivated promptly and caches are invalidated.
7. Owner and deputy access is reviewed separately by project management and IT.
8. All deployment, database, source-folder, and backup-folder sharing is included in the same review.

12. Environment, Deployment, and Release Management

12.1 Environment model

The Bound project supports TEST and PRODUCTION bindings through Apps Script Script Properties. TEST may fall back to the configured default database. PRODUCTION requires an explicit database ID and is validated through the core System Control service. The active environment, target database fingerprint, logical environment name, transaction mode, and core version can be inspected before use.

Property / control	Purpose
FMR_V3_ACTIVE_ENVIRONMENT	Selects TEST or PRODUCTION for the deployed Bound project.
FMR_V3_DATABASE_ID_TEST	Explicit test database override; optional because a default test database exists in source.
FMR_V3_DATABASE_ID_PRODUCTION	Required production spreadsheet identifier.
ENVIRONMENT_NAME	Logical database identity stored in Configuration; must agree with binding.
TRANSACTION_MODE	ENABLED permits writes; READ_ONLY blocks field/admin/owner data transactions.
Database fingerprint	Displays a non-raw identity for operator verification.
Library version	Bound manifest pins FMRCoreV3 version 19 for alpha.19.

12.2 Production deployment procedure

1. **Freeze the release candidate.** Tag or otherwise identify the exact commit and core library version. Do not deploy from an unreviewed working branch.
2. **Provision the production spreadsheet.** Create a dedicated production database with the exact required sheets/headers and approved ownership/sharing.
3. **Initialize system control.** Set project name, OWNER_EMAIL, environment name PRODUCTION, timezone, defaults, lists, and operational settings.
4. **Create production backup/source folders.** Restrict sharing and record ownership and retention responsibilities.
5. **Configure Bound Script Properties.** Set the production database ID and verify the active environment remains TEST during validation.
6. **Deploy the versioned core library.** Pin the reviewed library version in Bound/appsscript.json; do not use a development/head version for production.
7. **Run schema and contract diagnostics.** Resolve missing sheets, header mismatches, library incompatibility, environment mismatch, size-format issues, and operational-control failures.
8. **Execute acceptance fixtures in TEST.** Complete role, search, field, admin, staging, import, health, and recovery scenarios with evidence.
9. **Review deployment security.** Set the web-app audience, execute-as mode, framing policy, OAuth scopes, and Workspace sharing according to approved design.
10. **Take a pre-go-live backup.** Record the backup and validate it is accessible to the approved recovery owner.
11. **Activate PRODUCTION in READ_ONLY.** Inspect environment and database fingerprint; perform read-only validation first.
12. **Enable transactions.** Only after management and operational signoff, change transaction mode to ENABLED and communicate the go-live window.
13. **Monitor intensively.** Review health, errors, transaction counts, backorders, and user issues during the pilot period.

12.3 Release management minimum standard

Artifact	Required content
Release note	Version, commit, user-visible changes, schema/config changes, migrations, known issues, rollback plan.
Change approval	Business owner, technical owner, security/IT reviewer when material.
Test evidence	Diagnostic outputs and role-based acceptance results.
Deployment record	Date/time, deployer, environment, database fingerprint, library version, web-app deployment ID.
Backup evidence	Backup ID, completion time, retention, responsible owner.
Post-deployment check	Version, search, transaction gate, dashboard, health, and error review.
Rollback criteria	Conditions that trigger READ_ONLY, rollback, or restoration from backup.

Repository governance gap No README, license, contribution guide, changelog, or GitHub Actions workflow was observed in the reviewed tree. Add these before the repository becomes a shared operational asset.

13. Operations and Support Runbooks

13.1 Daily owner checklist

- Confirm the active environment and database fingerprint.
- Review overall health status and any FAIL/WARN component.
- Confirm the latest successful backup is within the approved age.
- Review pending and stale backorders.
- Review active and stale Bag & Tag records.
- Review failed imports, blocked items, or unresolved issues.
- Review unusual transaction volume or recent errors.
- Confirm scheduled operations trigger remains enabled and owned by the approved account.

13.2 Incident: suspected data inconsistency

1. **Protect the system.** Set transaction mode to READ_ONLY with a clear maintenance message.
2. **Record the incident.** Capture time, reporter, affected FMR/line, visible quantities, and recent action.
3. **Run health and diagnostics.** Check schema, integrity, indexes, notices, recent transactions, and backup age.
4. **Create a backup.** Preserve the current state before repair unless the backup action itself is unsafe.
5. **Preview recovery.** Use the narrowest action and inspect before/after scope.
6. **Apply approved recovery.** Refresh totals, rebuild index, sync notices, or safe reconcile as indicated.
7. **Validate.** Search the affected record, compare quantities to transactions, and run health again.
8. **Restore service.** Return to ENABLED only after technical and operational confirmation.
9. **Close the incident.** Record root cause, correction, impact, and preventive action.

13.3 Incident: unauthorized or wrong-role access

1. Deactivate the affected user immediately through System Control and invalidate access cache.
2. If owner or environment integrity is uncertain, set READ_ONLY and restrict deployment/Drive sharing.
3. Review Last_Login, Last_Interface, Audit_Log, Material_Transactions, and Apps Script logs.
4. Identify affected FMRs and compare transaction metadata and correlation IDs.
5. Correct quantities only through an approved recovery/correction path; do not edit rows silently.
6. Complete access-review and management notification before re-enabling the account.

13.4 Backup and recovery runbook

Stage	Procedure	Evidence
Routine backup	Allow the scheduled daily operation and verify SUCCESS status.	Backup history, file ID fingerprint, completed time.
Pre-change backup	Run manual backup before migration, large import, release, or recovery.	Notes reference the change/incident.
Retention	Confirm successful backups beyond the approved period are cleaned up.	Retention days and cleanup results.
Recovery preview	Select action and target; review before/after scope.	Recovery action PREVIEWED record.
Apply	Require reason and approved operator; system creates a recovery backup where designed.	Recovery APPLIED record and audit.
Validation	Run health, search, and affected workflow checks.	PASS result and operator signoff.
Rehearsal	Perform a scheduled recovery exercise in TEST.	Measured RTO/RPO and lessons learned.

13.5 Support triage

Symptom	First checks	Escalation
Cannot access	Signed-in email, active Users record, role, duplicate email rows.	Owner / Workspace admin
Search returns none	Exact FMR or ISO|sheet syntax, publication status, Search_Index health.	Owner / technical
Action not available	Line remaining, available/bagged/pending states, role, READ_ONLY mode.	Material Admin / Owner
Quantity error	Requested versus actual quantity, pending lock, bag source, current line state.	Material Admin
Backorder stuck	Queue status, pending quantity, operational index, admin permission.	Admin / Owner
Import blocked	Batch issues, worksheet format, required values, duplicates, limits.	Owner / technical
Health WARN/FAIL	Backup age, stale items, schema, system-control status, integrity details.	Owner / IT
Slow response	Apps Script quotas, cache, sheet size, lock contention, logging.	Technical / IT

14. Bulk Import Technical Guide

14.1 Source contract and limits

Control	Current alpha.19 behavior
Source types	Direct Excel upload or accessible Google Sheet.
Worksheet mapping	Each worksheet represents one proposed official FMR.
Maximum upload size	8 MB.
Maximum worksheets	100 per source.
Maximum material lines per FMR	35.
Maximum total lines	3,500 per batch.
Source fingerprint	SHA-256-derived fingerprint used for traceability/duplicate controls.
Parser version	ALPHA19_FRACTION_STORAGE_V1.
Size storage	All Size columns are enforced as plain text where sheets exist.
Output	Batch, item, line, and issue records; selected records become ordinary staging drafts.

14.2 Validation taxonomy

Category	Representative issue codes	Treatment
Structure	FORM_MARKER_MISSING, MATERIAL_HEADER_MISSING, MATERIAL_LINES_MISSING	Usually blocking; source layout does not meet parser expectations.
Header identity	OFFICIAL_FMR_MISSING, IWP_MISSING, ISO_NUMBER_MISSING, ISO_SHEET_MISSING	Blocking until corrected or an approved override applies.
Material line	COMMODITY_MISSING, SIZE_MISSING, DESCRIPTION_MISSING, QUANTITY_INVALID	Error or warning according to requiredness and parser policy.
Limits	LINE_LIMIT_EXCEEDED	Blocking; split or correct the source.
Duplicate/state	DUPLICATE_IN_BATCH, ALREADY_PUBLISHED, ALREADY_STAGED	Prevents accidental repeat publication/staging.
Optional metadata	REQUESTED_BY_BLANK, DELIVER_TO_BLANK, DATE_REQUIRED_BLANK, DESTINATION_BLANK, WAREHOUSE_BLANK, CRAFT_BLANK, REVISION_BLANK	Review and correct based on project policy.
Legacy content	HISTORICAL_ACTIVITY_IGNORED	Source may contain prior issue/backorder information that is intentionally not imported as live activity.
Spreadsheet coercion	SIZE_DATE_COERCION_REPAIRED, SIZE_DATE_COERCION_UNRESOLVED	Review fractional size values and source cell formatting.

14.3 Import design controls

- Uploaded files are tracked separately from published operational records.

- Batch status distinguishes parsing, review, valid, warning, blocked, staged, partially staged, superseded, and failed outcomes.
- Issue severity is recorded as ERROR, WARNING, or INFO, with resolution metadata.
- Existing staged/published FMR conditions are checked before staging.
- Historical activity in source forms is not silently converted into live material transactions.
- Owner selection is required; parsing alone does not publish records.
- Fractional sizes remain text across staging, published, bag, and import sheets to preserve construction-material notation.

14.4 Recommended source-preparation standard

- Use one official FMR per worksheet and a consistent worksheet naming convention.
- Preserve the approved form marker and material header labels.
- Enter official FMR number, IWP, ISO number, ISO sheet, revision, requester, date required, destination, warehouse, craft, and delivery contact where available.
- Keep each material line within the parser’s supported columns and line limit.
- Format size cells as text before entry; verify fractions after opening in Excel or Google Sheets.
- Remove hidden duplicate worksheets and obsolete drafts from the upload package.
- Do not rely on legacy issued/backordered columns to create operational history; use the live portal for transactions.

15. Testing, Diagnostics, and Quality Controls

15.1 Testing assets represented

Asset	Coverage
Schema diagnostics	Missing sheets and header mismatches.
Search diagnostics	Empty/known search, index lookups, phase timings and performance.
Admin diagnostics	Register, dashboard, active bag queue, operational rail and API contracts.
Bootstrap diagnostics	Version, user role, lists, UOM, backorder reasons.
Field acceptance fixture	Repeatable full field workflow and state transitions.
Admin acceptance fixture	Repeatable admin lifecycle and exception decisions.
Bound verification functions	Library compatibility, portal contracts, environment control, system control, operations, and feature-specific assertions.
Migration verification	Backfill candidate counts, orphan/inactive handling, duplicate deactivation, post-run verification.
Operational health	Schema, integrity, system control, backup freshness, stale exceptions/bags, recent transactions.
Production examples	Representative FMR PDFs/XLSX sources for parser development and validation.

15.2 Quality strengths

- Exact schema contracts fail loudly rather than accepting shifted spreadsheet columns.
- Critical operations use locks and validate state before writing.
- Publication tracks appended rows and original staging values to support rollback on failure.
- Indexes are active/inactive and invalidated after writes.
- Field actions return the updated serialized line so the UI can verify immediate state.
- Health, backup, and recovery are represented as operational data, not only console output.
- Bulk import uses fingerprints and a persistent issue register.
- Migrations verify the result and throw on failure.

15.3 Testing gaps and improvements

Gap	Risk	Recommended improvement
No conventional unit-test framework observed	Logic regressions may depend on manual Apps Script execution.	Extract pure state functions into testable modules and run automated tests.
No CI workflow observed	Repository changes may merge without lint, test, or manifest validation.	Add GitHub Actions with clasp-aware checks, ESLint, tests, schema snapshots, and secret scanning.
Diagnostics include hardcoded test identity/fixtures	Tests may be environment-specific or expose identifiers.	Move fixtures and identities to TEST configuration; prohibit production execution.
Acceptance evidence not stored as release artifact	A passing function today may not prove the deployed release passed.	Export diagnostic results per release and attach to deployment record.
No load/quota benchmark	Apps Script and Sheets limits may emerge with usage growth.	Define target concurrency/data volume and test search/write/import performance.
No independent security test	Authorization logic may not cover deployment or client risks.	Perform security review and targeted penetration test before broad rollout.

Gap	Risk	Recommended improvement
Large modules	Client, adapter, and bulk import code are difficult to review and maintain.	Refactor by feature/service and enforce module-size/code-review standards.

15.4 Minimum acceptance suite before production

Scenario	Expected evidence
Read Only user	Can search; all writes denied.
Field user	Can perform each valid action; cannot admin/owner.
Admin user	Can review backorders; cannot field/owner.
Owner user	Can stage, publish, configure users, run health/backup/recovery.
Environment mismatch	Production binding to TEST database is rejected or visibly fails validation.
READ_ONLY mode	Field/Admin/Owner data writes fail server-side with maintenance message.
Concurrent transaction	Second conflicting write is serialized/rejected without quantity corruption.
Backorder partial decision	Confirmed and pending portions remain mathematically consistent.
Bag partial issue	Tag and line remaining quantities reconcile.
Bulk import	Valid, warning, error, duplicate, fraction-size, line-limit and staging cases.
Backup/recovery	Backup succeeds; recovery preview and safe action validate; health returns PASS.
Audit	Representative actions can be traced by user, timestamp, entity, and correlation ID.

16. Risk and Readiness Assessment

16.1 Overall maturity rating

Maturity rating: Controlled pilot candidate The system has meaningful functional and operational controls, but the alpha release label and unresolved deployment, governance, testing, and platform-scale questions make a limited, monitored rollout more appropriate than unrestricted production adoption.

Dimension	Rating	Rationale
Business workflow	4 / 5	Strong alignment with FMR intake, field disposition, reservation, issuance, and exception handling.
Functional completeness	4 / 5	Broad features; external integrations and some operational conveniences remain future work.
Data controls	4 / 5	Strict schema, indexes, validation, locks, audit, state model, and recovery controls.
Security design	3 / 5	Good application roles and identity checks; deployment scope, framing, public repository, and hardcoded identifiers need review.
Reliability/continuity	3.5 / 5	Health, backups, scheduled operations, and recovery exist; formal rehearsal and support evidence are still needed.
Testing/release governance	2.5 / 5	Rich diagnostics/fixtures, but no conventional CI/test/release governance observed.
Scalability/maintainability	2.5 / 5	Indexes/caches help, but Apps Script/Sheets constraints and large source modules require management.
Production readiness	3 / 5	Appropriate for controlled pilot after P0 actions; not yet enterprise-grade without conditions.

16.2 Prioritized risk register

Priority	Risk	Likelihood	Impact	Mitigation
P0	Deployment exposure	High	High	Review ANYONE audience and ALLOWALL framing; restrict to approved Google Workspace audience.
P0	Production environment error	Medium	High	Dedicated production DB, explicit binding, logical environment match, fingerprint check, READ_ONLY first.
P0	Recovery not proven	Medium	High	Execute and document backup/recovery rehearsal and RTO/RPO.
P0	Access/Drive sharing	Medium	High	Formal user/folder/script sharing review; named owner/deputy; least privilege.
P0	Test evidence	Medium	High	Complete role, workflow, import, security-mode, concurrency, and continuity acceptance.

Priority	Risk	Likelihood	Impact	Mitigation
P1	Hardcoded environment/test values	High	Medium	Externalize IDs and identities; isolate fixture code from production.
P1	Platform quotas/contention	Medium	High	Measure data volume/concurrency; establish quota monitoring and escalation.
P1	No automated CI	High	Medium	Add lint/test/schema/secret checks and protected branch review.
P1	Large modules	High	Medium	Refactor Client, Adapter, BulkImport and services into reviewable modules.
P1	Public repository/binary artifacts	Medium	High	Assess confidentiality; remove sensitive examples; use releases/artifact storage.
P2	Manual spreadsheet editing	Medium	Medium	Protect database sheets/ranges and define approved break-glass procedures.
P2	Integration gap	Medium	Medium	Plan controlled exports/APIs to procurement, ERP, schedule, and reporting tools.
P2	Business continuity dependency	Low/Med	High	Document ownership transfer, Workspace account contingency, and platform exit strategy.

16.3 Production readiness conditions

- All P0 risks have approved evidence and owners.
- Production environment is distinct, access-controlled, backed up, and validated in READ_ONLY mode.
- Field, Admin, and Owner acceptance is signed by business representatives.
- Security/IT approve deployment audience, OAuth scopes, framing, Drive sharing, and owner contingency.
- Support, incident, change, backup, and recovery runbooks have named owners.
- Pilot scope, duration, KPIs, and rollback/stop conditions are approved.
- Alpha.19 is tagged or frozen as a release candidate, with known issues documented.

17. Recommended Roadmap and Management Decisions

17.1 30-day controlled production plan

Period	Actions	Exit criteria
Week 1 - Control closure	Security/deployment review; provision production DB/folders; externalize sensitive config; establish owner/deputy and support contacts.	P0 security and ownership approvals.
Week 2 - Validation	Run schema, role, workflow, import, concurrency, backup and recovery acceptance in TEST; remediate defects.	Signed test evidence and release candidate.
Week 3 - Limited launch	Deploy production READ_ONLY, validate identity; enable a small user group; monitor daily; log issues.	Stable health, correct transactions, no material incidents.
Week 4 - Pilot review	Measure KPIs, user feedback, error/rework, support burden, quota use, and control adherence.	Management stop/go/expand decision.

17.2 90-day improvement roadmap

Workstream	Priority deliverables
Security and governance	Restricted deployment, data classification, repository privacy decision, role-review cadence, ownership transfer, secrets/configuration standard.
Engineering quality	README, architecture decision records, changelog, release tags, protected branch, CI, lint, unit tests, secret scanning.
Maintainability	Split large modules, define internal APIs, centralize error handling, remove duplicate/legacy code and commented sprint patches.
Operations	Alerting for trigger/health/backup failures, quota dashboards, incident templates, monthly recovery exercise, capacity baselines.
User experience	Role-specific training, quick-reference cards, clearer action confirmations, optional barcode/tag workflow, export/reporting.
Integration	Prioritize ERP/procurement/document-control/schedule interfaces based on management value and system-of-record boundaries.
Analytics	Cycle time, fulfillment, aging, stale bags, exception reasons, import quality, adoption and productivity dashboards.

17.3 Management decisions requested

Decision	Recommended position
Approve controlled pilot?	Yes, contingent on P0 control closure and signed acceptance.
Declare enterprise production-ready?	No, not while release remains alpha and deployment/governance evidence is incomplete.
Use Google Sheets/Apps Script for the pilot?	Yes; it is cost-effective and aligned with current workflow, with documented capacity limits.
Keep repository public?	Reassess immediately. Use private access if examples, IDs, names, or future documents are confidential.

Decision	Recommended position
Name operating owner?	Yes. Assign a primary System Owner, approved deputy, business process owner, and IT escalation contact.
Fund engineering hardening?	Yes if the pilot demonstrates adoption and measurable operational benefit.
Define migration trigger?	Yes. Establish data volume, concurrency, integration, reliability, or compliance thresholds that would require a more traditional backend.

Final management recommendation Proceed with a controlled pilot, represent the system accurately as alpha.19, complete the production-control checklist, and use measured operational value to decide whether to harden the current Google Workspace architecture or migrate to a larger enterprise platform.

A. Glossary

Term	Definition
FMR	Field Material Request.
IWP	Installation Work Package.
ISO	Isometric drawing identifier; paired with sheet to form an operational search key.
Bag & Tag	A controlled reservation of physical material for an FMR/line.
Available	Physically located material not currently reserved or issued.
Backorder request	A Field-submitted exception awaiting Admin decision.
Confirmed backorder	Quantity accepted by Admin as unavailable/backordered.
Staging	Draft area used before publication.
Publication	Creation of live FMR header/line records and search indexes.
Correlation ID	Identifier linking related actions across transaction/audit/exception records.
Database fingerprint	Short non-raw identifier used to verify the active spreadsheet.
READ_ONLY	Maintenance mode that blocks data transactions.
Acceptance fixture	Repeatable test data and workflow used to validate a feature contract.
RTO	Recovery Time Objective.
RPO	Recovery Point Objective.

B. Configuration Reference

Setting	Default / allowed value	Purpose
PROJECT_NAME	FMR Operations v3	Portal/project identity.
ENVIRONMENT_NAME	TEST or PRODUCTION	Logical database environment.
TRANSACTION_MODE	ENABLED or READ_ONLY	Server-side write gate.
MAINTENANCE_MESSAGE	Blank unless READ_ONLY	Operator communication.
DEFAULT_CRAFT	PIPE	New FMR default.
DEFAULT_DESTINATION	Field	New FMR default.
DEFAULT_DELIVER_TO	Configured project delivery contact	New FMR default; review before production.
DEFAULT_PRIORITY	Normal	New FMR default.
TIMEZONE	America/Indiana/Indianapolis	Timestamp formatting and schedule context.
BACKUP_RETENTION_DAYS	30	Successful backup retention.
BACKUP_MAX_AGE_HOURS	30	Backup freshness warning threshold.
STALE_BACKORDER_HOURS	48	Pending-backorder warning threshold.
STALE_BAG_DAYS	7	Active reservation warning threshold.

Setting	Default / allowed value	Purpose
HEALTH_HISTORY_LIMIT	100	Maximum retained health records.
SEARCH_CACHE_SECONDS	Configured; fallback 3600	Index cache TTL within enforced bounds.
SEARCH_INDEX_VERSION	Configured; fallback 1	Index cache/version key.

C. Repository File Inventory

Area	Files / artifacts
Bound	Adapter.gs; Client.html; FieldWorkflowStyles.html; Index.html; Styles.html; WebApp.gs; appsscript.json
FMRCoreV3 - foundation	Config.gs; Repository.gs; IndexServiceFmr.gs; Security.gs; PublicApi.gs; AuditService.gs
FMRCoreV3 - business services	SearchService.gs; FieldService.gs; FieldMetadataService.gs; FieldBackorderNoticeService.gs; BackorderService.gs; AdminRegisterService.gs; AdminActiveBagService.gs; DashboardService.gs; OwnerService.gs
FMRCoreV3 - control services	SystemControlService.gs; OperationalReadinessService.gs; IntegrityService.gs; Migrations.gs; Diagnostics.gs
FMRCoreV3 - intake/testing	BulkImportService.gs; FieldAcceptanceFixtureService.gs; AdminAcceptanceFixtureService.gs; appsscript.json
Documentation	Prior technical DOCX and PDF in Documentation/.
Examples	Representative production FMR PDFs and XLSX workbooks in ProductionFMRExamples/.
Repository hygiene	.gitignore; historical release ZIP artifacts are committed at repository root.

C.1 Repository observations

- The current source snapshot is approximately 42 commits ahead of the initial commit and reached alpha.19 within a rapid development cycle.
- The repository contains both source and binary release/document/example artifacts.
- No README.md was present at the reviewed main-branch path.
- No GitHub Actions workflow was observed in the reconstructed reviewed tree.
- The existing documentation artifact name indicates coverage through a prior “Production Readiness” sprint and should be treated as historical unless reconciled with alpha.19.
- Large source files increase review and maintenance cost; modularization should be included in hardening.

D. Go-Live Checklist

Checklist area	Completion requirement
Governance	Named business owner, System Owner, deputy, IT/security contact, support channel, change approvers.
Release	Frozen commit/library version, release note, known issues, rollback plan.
Production database	Dedicated spreadsheet, exact schema, PRODUCTION identity, protected sharing, fingerprint recorded.
Deployment	Approved audience, execute-as policy, framing policy, OAuth scope review, deployment ID recorded.

Checklist area	Completion requirement
Access	Users reviewed, least privilege, owner/deputy tested, deactivation tested.
Configuration	Project defaults, lists, timezone, maintenance message, thresholds, folders.
Testing	Schema, contract, role, field, admin, owner, import, concurrency, read-only, backup, recovery acceptance.
Data	Approved initial FMR/import package; no confidential test data; source files retained appropriately.
Continuity	Successful pre-go-live backup, folder access, recovery rehearsal, RTO/RPO.
Operations	Daily schedule installed, health/backup monitoring, incident runbook, escalation.
Training	Role-specific training and quick-reference instructions completed.
Pilot	Scope, users, duration, KPIs, review date, expansion/rollback criteria approved.

E. Source Basis and Review Limitations

This document was prepared from the public GitHub repository jonaDaniM/FMRv3, main branch, current source version 3.0.0-alpha.19, with the latest reviewed commit identified as 52d5f902e1e70dd1477dbe93e348a06eec2a3d04 (“Phase 19 implemented”). The review covered repository metadata, commit history, current Apps Script manifests, Bound web-app files, FMRCoreV3 configuration and public API, security, repository/index layers, Field/Admin/Owner workflows, bulk import, operational readiness, diagnostics, migrations, and repository file inventory.

The review did not connect to the deployed web app, inspect the live production/test spreadsheets, verify Google Workspace sharing, execute OAuth consent, validate Apps Script quotas, run the acceptance fixtures, or inspect cloud logs. Statements about implementation are based on code paths and manifests. Statements about readiness are professional assessments, not a certification or warranty.

Evidence area	Primary repository sources
Repository and release	Repository metadata; initial and latest commit records; initial-to-latest file comparison.
Runtime/deployment	Bound/appsscript.json; FMRCoreV3/appsscript.json; Bound/WebApp.gs; Bound/Adapter.gs.
Architecture/API	FMRCoreV3/PublicApi.gs; Config.gs; Repository.gs; IndexServiceFmr.gs.
Security/control	Security.gs; SystemControlService.gs; AuditService.gs.
Business workflows	SearchService.gs; FieldService.gs; BackorderService.gs; AdminRegisterService.gs; OwnerService.gs.
Bulk intake	BulkImportService.gs and production example file inventory.
Operations/quality	OperationalReadinessService.gs; Diagnostics.gs; Migrations.gs; acceptance fixture module inventory.

> **Document maintenance** Update this documentation whenever the library version, schema, role model, environment binding, deployment manifest, critical workflow, or operational thresholds change.